4-2 POINT ESTIMATION

- The fundamental goal of statistical estimation:

  - **Definition**: The primary objective is to select a single, plausible number based on random sample data to represent an unknown population parameter.

  - **Notation**: The Greek letter θ **(theta)** is used as a general symbol to represent any parameter of interest, such as the population mean or variance.

  - **Application**: A sample statistic is calculated to provide these **point estimates**.

- **Point Estimator**: A statistic $\hat{\Theta}$ (calculated as a function h of a random sample) used to approximate an unknown population parameter θ. Because it is a function of random variables, the **point estimator** is itself a random variable.

- **Point Estimate**: The specific numerical value that the **point estimator** produces once an actual sample has been collected and calculated.

A **Point Estimate** is a single numerical value used to approximate a specific population parameter.

- **Definition**: It is a specific value calculated from sample data.
- **Purpose**: A **point estimate** serves as a "best guess" or single-number approximation of an unknown population characteristic (like a mean or proportion).
- **Representation**: It is represented by a statistic, often denoted as $\hat{\theta}$.

- Engineering often requires making estimations about various population parameters. The key areas include:

  - **Single Population Metrics:** Estimating the average (**mean**), the spread (**variance/standard deviation**), or the frequency of a specific trait (**proportion**).

  - **Comparative Metrics:** Determining the gap between two different groups by calculating the **difference in means** or the **difference in proportions**.

The point estimates for various population parameters are:

- **Population Mean (μ):** Estimated by the **sample mean** ($\bar{x}$).
- **Population Variance (σ^2):** Estimated by the **sample variance** (s^2).
- **Population Proportion (p):** Estimated by the **sample proportion** ($\hat{p} = x/n$).
- **Difference Between Two Means ($\mu_1 - \mu_2$):** Estimated by the **difference between two independent sample means** ($\bar{x}_1 - \bar{x}_2$).

- **Difference Between Two Proportions ($p_1 - p_2$):** Estimated by the **difference between two independent sample proportions** ($\hat{p}_1 - \hat{p}_2$).

- The following defines the core concept of an **Unbiased Estimator** in statistics:

  - **Unbiased Estimator**: A point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if its expected value is equal to that parameter. This is shown in:
    E(Θˆ) = θ (4-1)

  - **Bias**: If the estimator is not unbiased, the resulting difference is called the bias of the estimator $\hat{\Theta}$. This is expressed as:
    E(Θˆ) - θ (4-2)

EXAMPLE 4-1 Unbiased Estimators

- **Goal**: Demonstrate that the sample mean ($\bar{X}$) and sample variance (S^2) are unbiased estimators of the population mean (μ) and population variance (σ^2).

- **Context**: Based on a random sample $X_1, X_2, \ldots, X_n$ from a population with mean μ and variance σ^2.

Solution

- **Sample Mean**: Since $E(\bar{X}) = \mu$, $\bar{X}$ is confirmed as an unbiased estimator of μ.
- **Sample Variance**: Since $E(S^2) = \sigma^2$, S^2 is confirmed as an unbiased estimator of σ^2.
- **Important Note**: While S^2 is unbiased, the sample standard deviation (S) is a **biased** estimator of σ, though this bias becomes negligible in large samples.

- Key points regarding **Figure 4-2** and the comparison of unbiased estimators:

  - **Unbiasedness**: Both $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are unbiased, meaning their distributions are centered exactly at the true value of θ.

  - **Variance Comparison**: While both are centered correctly, they differ in their spread. **Figure 4-2** shows that $\hat{\Theta}_1$ has a smaller variance than $\hat{\Theta}_2$.

  - **Precision**: Because it has a smaller variance, $\hat{\Theta}_1$ is more likely to yield an estimate closer to the actual value of θ.

- **Definition**: Among all possible unbiased estimators for a parameter θ, the specific one that maintains the lowest variance is known as the **Minimum Variance Unbiased Estimator** (MVUE).

- **Efficiency**: The **minimum variance unbiased estimator** is considered the most "efficient" choice because it consistently yields estimates closest to the true value with the least amount of scatter.

- **Core Concepts**: Unbiasedness and minimum variance are the primary criteria for evaluating estimators.
- **Formal Derivation**: Specific statistical methods exist to derive parameter estimates for probability distributions.
- **The method of maximum likelihood**: This particular approach yields point estimators that are nearly unbiased and approach the efficiency of a minimum variance estimator.
- **Reference**: Montgomery and Runger (2011) is the recommended source for an in-depth look at this method.

- When using a biased estimator (like S for σ), the **mean square error** serves as a key metric for evaluation.

  - **Definition**: The **mean square error** of an estimator $\hat{\Theta}$ is the expected squared difference between the estimator and the true parameter θ.

  - **Application**: It is particularly useful for assessing the performance of estimators that do not center exactly on the target value.

- **Definition:** The **Mean Square Error of an Estimator** represents the expected value of the squared difference between an estimator $\hat{\Theta}$ and the actual parameter θ.

- **Formula:** The mathematical expression for the **mean square error** is given by:

$$MSE(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2 \quad \textbf{(4-3)}$$

- The Mean Square Error (MSE) measures an estimator's total error by combining its precision (**variance**) and accuracy (**bias**).

Key Components

- **Variance ($\mathrm{Var}(\hat{\Theta})$):** The "spread" of the estimator; how much it fluctuates around its own average.
- **Bias ($E[\hat{\Theta}] - \theta$):** The systematic error; how far the average estimate falls from the true value.

The Equation

The relationship is defined by the **bias-variance decomposition**:

$$MSE(\hat{\Theta}) = \mathrm{Var}(\hat{\Theta}) + [\mathrm{Bias}(\hat{\Theta})]^2$$

Summary Points

- **The Total Error:** MSE is equal to the variance plus the square of the bias.
- **Unbiased Estimators:** If an estimator is unbiased (Bias = 0), the MSE is simply equal to the **variance**.
- **The Trade-off:** A "good" estimator aims to minimize both components, though reducing one often increases the other.

- Defines a method for comparing the quality of two statistical estimators using their Mean Square Errors (MSE).

  - **Core Criterion**: The Mean Square Error (MSE) is the primary tool used to compare the performance of two different estimators, $\hat{\Theta}_1$ and $\hat{\Theta}_2$.
  - **Definition**: The ratio of the two mean square errors is known as the **relative efficiency**.
  - **Formula**: The calculation is expressed as $MSE(\hat{\Theta}_1)/MSE(\hat{\Theta}_2)$ **(4-4)**.
  - **Interpretation**: If the result of the **relative efficiency** formula is less than 1, $\hat{\Theta}_1$ is considered the superior (more efficient) estimator because it produces a smaller error.

- **Standard Error**: This represents the standard deviation of a statistic's sampling distribution.

- **estimated standard error**: This is created when unknown parameters within the **standard error** are replaced with their estimated values.

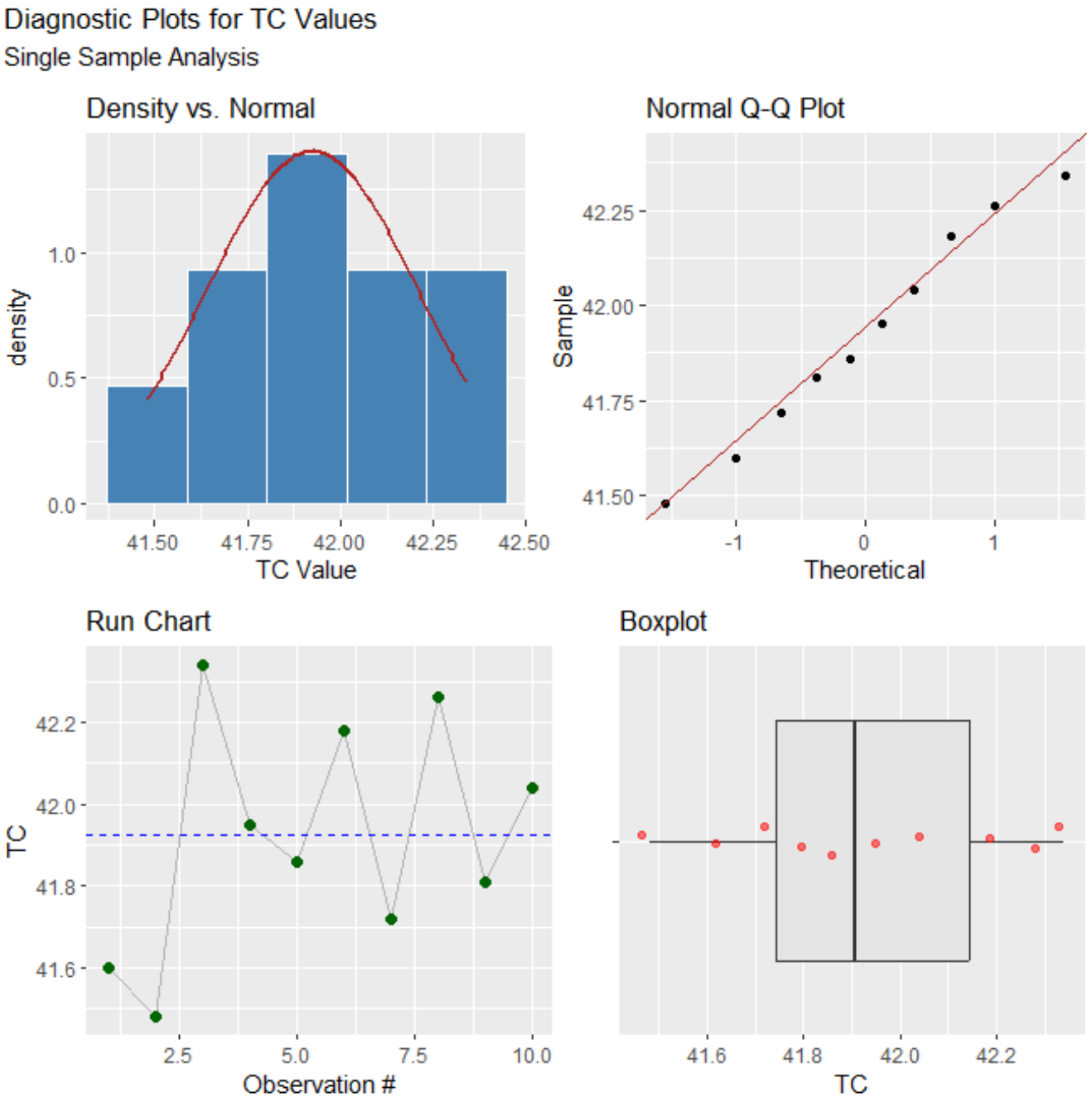
- An article in the **Journal of Heat Transfer** (**Trans. ASME**, 1974) determined the thermal conductivity of Armco iron at 100°F and 550 watts using 10 measurements (41.60, 41.48, 42.34, 41.95, 41.86, 42.18, 41.72, 42.26, 41.81, 42.04 Btu/hr-ft-°F). The study calculated a sample mean of 41.924 and an estimated standard error of 0.0898, indicating high precision [1].

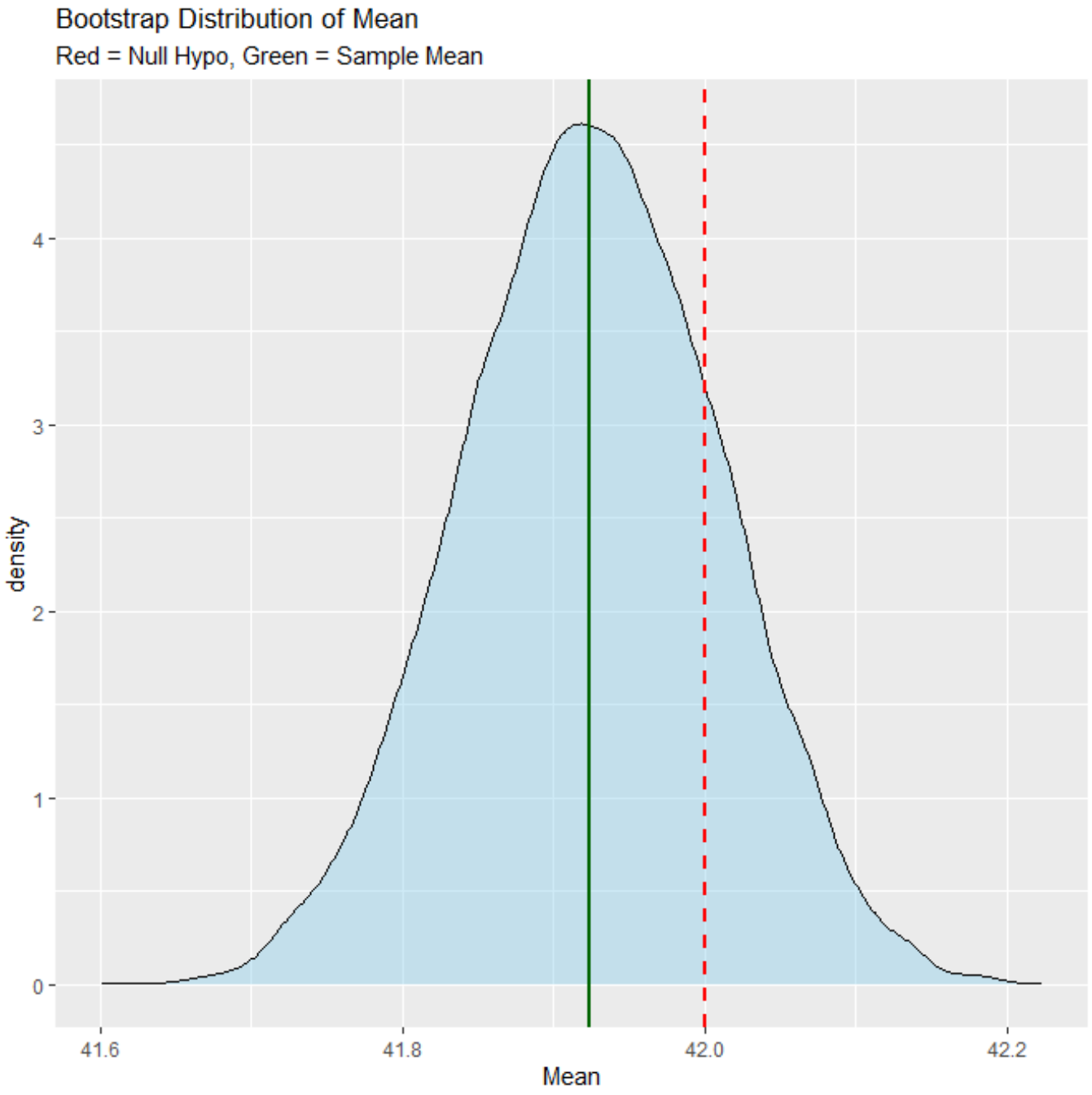
Key Details:

- **Conditions:** 100°F at 550 watts.
- **Data Set (Btu/hr-ft-°F):** 41.60, 41.48, 42.34, 41.95, 41.86, 42.18, 41.72, 42.26, 41.81, 42.04.

- **Sample Mean ($\bar{x}$):** 41.924 Btu/hr-ft-°F.
- **Sample Standard Deviation (s):** 0.284.
- **Calculating the Standard Error:** $\hat{\sigma}_{\bar{x}} = s/\sqrt{n} = 0.284/\sqrt{10} = 0.0898$

  .
- **Result:** The standard error is approximately 0.2% of the sample mean.


Diagnostic Plots for TC Values
Single Sample Analysis
Density vs. Normal
density
1.0
0.5
0.0
41.50
41.75
42.00
42.25
42.50
TC Value
Normal Q-Q Plot
Sample
42.25
42.00
41.75
41.50
-1
0
1
Theoretical
Run Chart
TC
42.2
42.0
41.8
41.6
2.5
5.0
7.5
10.0
Observation #
Boxplot
41.6
41.8
42.0
42.2
TC


Bootstrap Distribution of Mean
Red = Null Hypo, Green = Sample Mean
density
0
1
2
3
4
41.6
41.8
42.0
42.2
Mean